

Are galaxies in compact groups special?

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ABSTRACT

Context. Compact groups combine the highest projected galaxy densities with low velocity dispersions, yet most of them lie inside larger groups. Whether their galaxies are special therefore depends on how they compare with the members of ordinary groups, not only with the field.

Aims. We test whether the galaxies of compact groups of four members (CG₄) differ from those of regular groups of four members (RG₄), from the four brightest members of regular groups (Control_{4B}), and from the projected cores formed by the brightest group galaxy (BGG) and its three nearest projected companions (Control_{4C}).

Methods. For 62 non-split CG₄ systems we combine Galaxy Zoo 1 morphologies, stellar masses, specific star-formation rates, emission-line diagnostics, and seeing-corrected half-light radii, and compare each control separately with covariate-adjusted models that include the satellites' projected distance to their BGG.

Results. The clearest difference is morphological. At fixed stellar mass, redshift, and group properties, CG₄ satellites are more often in the Galaxy Zoo 1 E class than control satellites (odds ratios 2.63, 1.93, and 3.03 against Control_{4B}, Control_{4C}, and RG₄), whereas BGGs show no comparable excess. These satellites lie much closer to their BGG in projection, where E-class galaxies are more common in all samples; at the masses and projected distances of the CG₄ satellites, the controls predict E-class fractions of 0.43, 0.49, and 0.42, against 0.56 observed. At fixed morphology and stellar mass, CG₄ satellites are more often quenched than Control_{4B} satellites, but not detectably more than Control_{4C} or RG₄ satellites. The deficit of strong H α emission in CG₄ lies in high-[N II]/H α rather than star-forming-like emission. Sizes, colours, and magnitude gaps show no comparably robust offsets.

Conclusions. Compact-group satellites are shifted towards early-type-like morphologies; the shift is reduced, but persists, at fixed projected distance from their BGG. Evidence for additional quenching is weaker and depends on the comparison sample.

Key words. galaxies: groups: general – galaxies: evolution – galaxies: interactions – galaxies: star formation – galaxies: elliptical and lenticular, cD – methods: statistical

1. Introduction

Given their high apparent densities and low velocity dispersions (Hickson 1982; Hickson et al. 1992), compact groups appear to be ideal sites for the frequent tidal encounters and mergers that alter galaxy masses, luminosities, sizes, morphologies, and star-formation rates (e.g., Mamon 1992). Mendes de Oliveira & Hickson (1994) provided the classic report of an excess of early-type galaxies in compact groups relative to the field (building on the photometric morphologies of Hickson et al. 1989 and the merger-remnant expectations of Zepf & Whitmore 1993), in line with the broader morphology–density relation (Dressler 1980); multi-wavelength studies of compact-group galaxies likewise report altered infrared, UV, star-formation, and stellar-population properties relative to looser environments (Johnson et al. 2007; Tzanavaris et al. 2010; Bitsakis et al. 2011; Coenda et al. 2012), and their H I de-

ficiency has been organised into an evolutionary sequence (Verdes-Montenegro et al. 2001).

The interpretation of such contrasts is nevertheless ambiguous. Projected selection can include chance alignments, and many Hickson-like systems need not be isolated bound entities: apparent compact groups may be transient alignments, dense substructures, or cores embedded in looser groups and richer haloes (Mamon 1986; Díaz-Giménez & Mamon 2010; Díaz-Giménez et al. 2020; Zheng & Shen 2021; Taverna et al. 2023). The relevant observational question is therefore not only whether compact-group galaxies differ from field galaxies, but whether they remain different from galaxies occupying regular-group analogues selected with comparable luminosity, redshift, and membership constraints. Recent structural studies find evidence for transformation in compact groups, especially non-isolated systems (Montaguth et al. 2023, 2025, 2026a,b); Montaguth et al. (2026b) compared non-isolated compact-group galaxies with surrounding host populations, but not with ordinary-group quartets reproducing compact-group

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membership, luminosity-ranking, or projected-core selection.

In our previous work (Tricottet et al. 2025), we found that, once the direct consequences of the compact-group selection are accounted for, compact groups of four galaxies have group-scale properties broadly similar to those of regular groups of four galaxies and to the eligible cores of regular groups. Here we ask how the galaxy-population contrast changes across three ordinary-group comparison samples: genuine four-member groups (RG₄), the four brightest eligible members of richer groups (Control_{4B}), and BGG-centred projected cores (Control_{4C}), the closest analogue of a compact configuration. We adjust for stellar mass, redshift, BGG or satellite status, and group-scale properties, ask how much of each contrast is associated with where the galaxies sit within their groups, and compare compact-group members with the other members of their own host groups. Section 2 describes the samples, the classifications, and the statistical approach; Sect. 3 presents the results; Sect. 4 discusses their physical interpretation; and Sect. 5 concludes.

2. Data and classifications

2.1. Samples

We use the compact-group and control samples constructed in our previous study (Tricottet et al. 2025). Galaxy properties are taken from the SDSS DR12 catalogue of Tempel et al. (2017), while regular groups are drawn from the SDSS DR7 group catalogue of Lim et al. (2017); the two catalogues were matched through SDSS object identifiers to define the Lim–Tempel working catalogue. Compact groups are selected from the modified Hickson compact-group catalogue of Díaz-Giménez et al. (2018): reliable HMCs with exactly four members, all present in the Lim–Tempel catalogue, satisfying the volume- and luminosity-complete limits $0.005 \leq z_{\text{BGG}} \leq 0.0452$ and $M_{r,\text{BGG}} \leq -21.81$, with all four galaxies lying within 3 magnitudes of the brightest group galaxy (BGG). In contrast to our previous study, we remove CGs classified as *split* following Zheng & Shen (2021), which reduces the compact-group sample from 78 to 62 groups. Following Zheng & Shen (2021), the remaining systems are *isolated* when the four members constitute their whole Lim–Tempel group, *pre-dominant* when they contribute at least half of the host-group *r*-band luminosity, and *embedded* otherwise (6, 37, and 19 systems).

The three control samples are built from Lim–Tempel groups satisfying the same redshift and BGG-luminosity limits, with at least four members and at least three satellites within 3 magnitudes of the BGG, and serve different purposes (Fig. 1). RG₄ contains regular groups with exactly four members and is the most direct equal-membership comparison. Control_{4B} contains the four brightest eligible members of each regular group and tests whether the CG₄ population resembles the luminous core of an ordinary group. Control_{4C} contains the BGG and its three closest projected companions among members within 3 magnitudes of the BGG, reconstructed here from the eligible parent groups by applying the magnitude criterion before ranking companions by projected distance; it is the most compact-core-like analogue, but not a compactness-matched sample: the CG₄–Control_{4C} overlap coefficient in

Table 1. Final samples

Sample	Groups	Galaxies	$\langle R_{ij} \rangle$ (kpc)
CG ₄	62	248	143 (114–177)
Control _{4B}	698	2792	447 (323–604)
Control _{4C}	703	2812	293 (187–436)
RG ₄	56	224	430 (292–494)

Notes. $\langle R_{ij} \rangle$ is the median of the six pairwise projected physical separations in each quartet, derived from angular-diameter distances in the adopted Planck 2015 cosmology. Entries give the median and interquartile range over groups.

$-\log\langle R_{ij} \rangle$ is 0.34, and 18% of CG₄ groups lie above the Control_{4C} 95th percentile in compactness. Every control group containing at least one CG₄ galaxy is excluded, so no galaxy appears on both sides of any comparison; Appendix C shows that this exclusion removes the hosts of 51 of the 56 non-isolated compact groups, whose members form the would-be Control_{4C} cores of those hosts. We also remove the systems associated with Lim group 3688, whose single outlying-redshift member ($z = 0.048$, against 0.032–0.034 for the other six) enters both the Control_{4B} and Control_{4C} quartets and inflates their velocity dispersions to $\approx 2000 \text{ km s}^{-1}$. Table 1 summarises the final samples; the Control_{4B} and Control_{4C} group counts differ because the luminosity-ranked and distance-ranked definitions retain different Lim groups after the compact-group exclusion.

We use SDSS DR16 (Ahumada et al. 2020) to retrieve stellar masses, star formation rates (SFRs; Brinchmann et al. 2004; Kauffmann et al. 2003b), emission-line measurements, and Galaxy Zoo morphologies (Lintott et al. 2008, 2011); Appendix A lists the tables and joins. Throughout, ‘SDSS non-AGN reference’ denotes the AGN-filtered reference sample used for calibration and orientation: spectroscopic galaxies with $0.005 \leq z \leq 0.0452$, extinction-corrected Petrosian $r \leq 17.77$, a valid MPA-JHU stellar mass, and finite Galaxy Zoo vote fractions after the AGN filter. It is a field-dominated reference population, not a control sample. Projected physical separations are derived from angular-diameter distances in the adopted Planck 2015 cosmology (Planck Collaboration et al. 2016), as in Paper I; all distances quoted below are in kpc.

2.2. Classifications

Star-formation classes are based on SDSS total specific star-formation rates. The inherited DR12 group catalogues store $\log \text{sSFR} = \text{sfr_tot_p50} - \text{l gm_tot_p50}$. In the MPA-JHU catalogues, -9999 is the missing-value sentinel of the `gal-SpecExtra` table (Stoughton et al. 2002; Brinchmann et al. 2004; Kauffmann et al. 2003b), so rows with unavailable components are treated as *missing data*: they are excluded from the classification and from every sSFR-based figure and fraction, and are reported as counts only (Table 2). All quenched and star-forming fractions are computed among classified galaxies. The missing-sSFR incidence is higher in CG₄ (7% of members, 10% of BGGs) than in the reference sample (1%). Tracing the stored spectrum identifiers shows why: 332 of the 394 missing rows are BOSS-instrument spectra, which the MPA-JHU tables (legacy reductions only) never covered, 60 are legacy spectra whose SFR fit re-

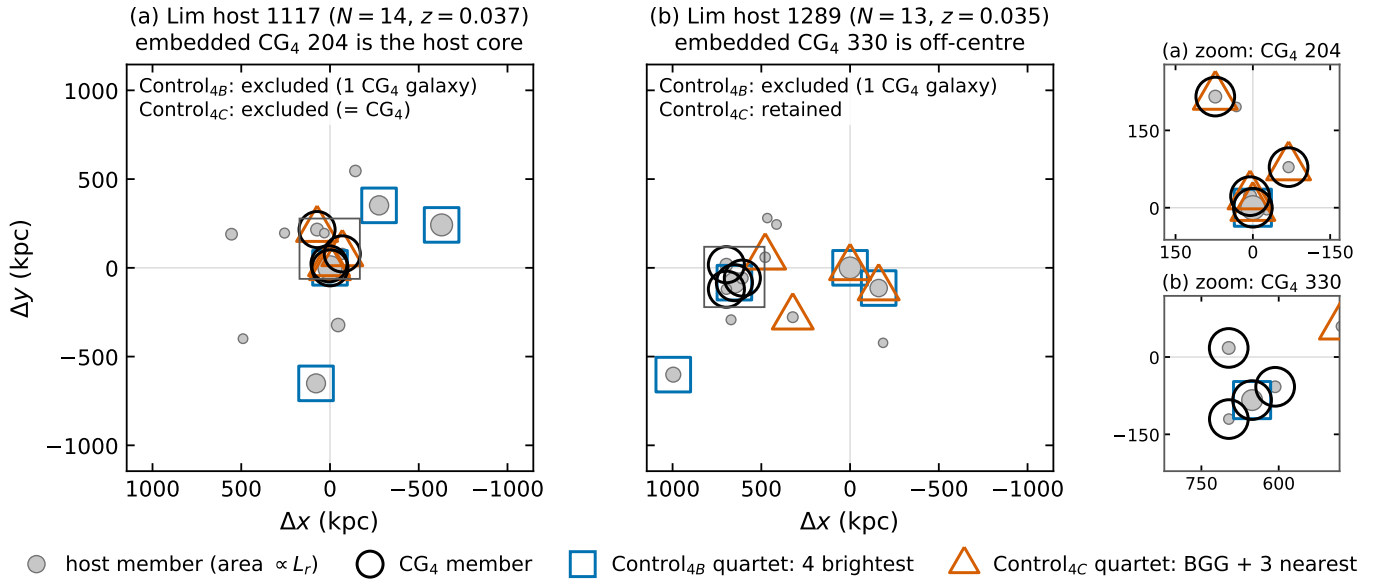


Fig. 1. Sample definitions on the sky: two Lim–Tempel groups hosting embedded compact groups, in projected physical coordinates relative to the host BGG (east to the left, common scale); disc area is proportional to r -band luminosity. Black circles mark the CG₄ members, blue squares the four brightest members within 3 mag of the BGG (the Control_{4B} quartet), and orange triangles the BGG and its three nearest projected companions within 3 mag (the Control_{4C} quartet); the right-hand panels enlarge the boxed ± 170 kpc windows around the compact groups. (a) CG₄ 204 is the would-be Control_{4C} core of Lim group 1117 and contains its BGG, so both quartets are excluded from the controls, as for 51 of the 56 non-isolated compact groups (Appendix C). (b) CG₄ 330 is an off-centre substructure 670 kpc from the BGG of Lim group 1289: the Control_{4C} quartet contains no compact-group galaxy and is retained, whereas the Control_{4B} quartet contains the brightest CG₄ member and is excluded.

turned the sentinel, and 2 have no spectrum identifier; the missing rows have higher, not lower, median signal-to-noise than the classified ones. They are, however, concentrated among galaxies with a close neighbour: 28% of the galaxies within 55 arcsec have no sSFR, against 3% of the others (Appendix A). Galaxies with a measured sSFR are split into quenched and star-forming classes along the valley between the two populations in the $\log M_\star$ – $\log$ sSFR plane of the reference sample, located with a two-component Gaussian mixture fitted to its binned distribution; the fit, the AGN handling, and alternative classifications are described in Appendix B (Fig. B.1).

Morphologies are drawn from Galaxy Zoo 1 (Lintott et al. 2008, 2011), whose debiased catalogue vote fractions (Bamford et al. 2009) are `p_el_debiased`, labelled “elliptical” in GZ1, and `p_cs_debiased`, labelled combined spiral. We define the GZ1 E class by `p_el_debiased` > 0.5, the S class by `p_cs_debiased` > 0.5, and the uncertain class when both vote fractions are finite but neither threshold is exceeded; rows with a missing vote fraction are treated as missing Galaxy Zoo data. GZ1 does not separate E from S0, so throughout this paper the E class is interpreted as a smooth early-type-like class, not as a physical sample of classical ellipticals. A cross-match with the deep-learning morphologies of Domínguez Sánchez et al. (2018) (DS18; 2436 of 3857 unique galaxies) quantifies this: of the 942 matched E-class galaxies, 92% have DS18 $T\text{Type} \leq 0$, 54% are early with $P_{S0} > 0.5$, and 29% with $P_{S0} \geq 0.8$; conversely, DS18 early types with $P_{S0} > 0.5$ ($N = 730$) fall 70% in our E class and 19% in S (63% and 24% for $P_{S0} \geq 0.8$), with the E share dropping from 77% for face-on to 47% for inclined systems

(DS18 $P_{\text{edge-on}} < 0.1$ versus ≥ 0.5), as expected if inclined S0s expose their discs. We reserve “elliptical” for the literal GZ1 category and for external classifications that physically distinguish ellipticals. For the stored spectra of the group galaxies we also retrieve the MPA-JHU age-sensitive indices D_n4000 and $H\delta_A$ (Kauffmann et al. 2003b), used descriptively (Sect. 3.4), and the fluxes and equivalent widths of $H\alpha$, $[\text{N II}]\lambda 6584$, $H\beta$, and $[\text{O III}]\lambda 5007$ with their uncertainties (Appendix A).

2.3. Galaxy sizes

Our primary size measure is the circular effective (half-light) radius of the pure Sérsic fit in the r band, $R_{e,r}$, from the structural catalogue of Simard et al. (2011): PSF-convolved GIM2D fits with re-derived sky levels and deblending designed for crowded fields, so the sizes are corrected for seeing. The catalogue is keyed to SDSS DR7 identifiers; we bridge our DR16 objects to their DR7 counterparts through the SkyServer SpecDR7 cross-match table. Rows whose Simard redshift differs from the catalogue redshift by more than 0.005 are treated as wrong matches and dropped (9 rows). The Simard sizes, tabulated in kpc under the catalogue’s own cosmology, are converted back to angular sizes with the catalogue plate scale and then to kpc with the same Planck 2015 conversion as every other projected quantity.

As a secondary, model-independent measure we use the SDSS Petrosian radii `petroR50_r` and `petroR90_r`, which also define the concentration index $C = R_{90,r}/R_{50,r}$ and provide the like-for-like bridge to Coenda et al. (2012). Petrosian radii are not corrected for seeing, and the Petrosian aperture misses roughly 20% of the flux of an $n = 4$

profile, which biases $R_{50,r}$ low for concentrated galaxies (Blanton et al. 2001; Graham et al. 2005). All sizes are required to lie between 0.1 and 50 kpc after re-conversion; Sérsic fits pegged at the GIM2D bounds ($n_g \leq 0.55$ or $n_g \geq 7.9$; 60 galaxies) are excluded from the primary sample, a cut that removes 2.5% of the Simard-matched CG₄ members against 1.0–1.5% in the controls and is therefore conservative with respect to a CG₄ early-type excess; Petrosian entries must satisfy $R_{50,r} > 0$, $R_{90,r} > R_{50,r}$, and a finite positive uncertainty. The completeness of both measures is given in Appendix G.4.

2.4. Statistical approach

Our main question is whether compact-group galaxies retain distinct properties once differences in stellar mass, redshift, BGG or satellite status, and group properties are taken into account. The three controls address physically different comparisons, so we compare CG₄ with each one separately. For each control, binomial logistic models fitted to the galaxies, or to the satellites only, of CG₄ and that control quantify the adjusted E-versus-S and quenched-versus-star-forming contrasts. The covariates are stellar mass, redshift, a BGG/satellite indicator, the total r -band luminosity of the quartet, and the quartet velocity dispersion σ_v ; because σ_v from four tracers is noisy, every analysis that uses it is repeated without it (Appendix H). Odds ratios from models with different covariates are not directly comparable, so for the key contrasts we also give the fraction expected for galaxies with the covariate values of the CG₄ galaxies if they followed the relation of the control sample, to be compared with the observed CG₄ fraction (Appendix D). Within-host models compare compact-group members with their non-compact co-members, and one-to-one matching of galaxies and of quartets is used as a robustness check (Appendix G).

The independence unit is the *physical* Lim group: standard errors are clustered by physical group, and bootstrap intervals resample physical groups, separately within CG₄ and the control, or matched pairs, because the four galaxies of one group are not independent. We report two-sided p-values with effect sizes and confidence intervals for the planned morphology, star-formation, host, and size analyses, and reserve multiplicity control for the exploratory batteries. The p-value conventions, the covariate set, the overlap coefficient, and the treatment of the overlapping control samples in pooled summaries are collected in Appendix D.

3. Results

3.1. Morphology

CG₄ galaxies are more often in the GZ1 E class than the galaxies of every control. Tables 2 and 3 give the raw class counts; among galaxies with a usable E or S classification, the CG₄ E-class fraction is 59% against 47%, 48%, and 32% in Control_{4B}, Control_{4C}, and RG₄ (Fisher exact tests: $p = 6.8 \times 10^{-4}$, $p = 1.9 \times 10^{-3}$, and $p < 10^{-4}$). The 38 CG₄ rows with uncertain or missing Galaxy Zoo classifications are excluded rather than assigned to either class. Satellites account for most of the difference: among usable BGG classifications the E-class fraction is 67.9% in CG₄ and 44.7% in RG₄ (Fisher $p = 1.00$, 1.00, and 0.03 against Control_{4B},

Control_{4C}, and RG₄), and restricted to star-forming galaxies the E/S mix is compatible with every control (two-sided Fisher exact test: $p = 0.332$, $p = 0.468$, $p = 0.226$), so star-forming E-class galaxies do not produce the raw excess.

The mass-binned view separates BGGs from satellites and shows that the E-class excess is primarily a satellite result rather than a stellar-mass artefact (Fig. 2). The quenched fractions do not show an analogous CG₄-specific pattern.

The excess survives adjustment for stellar mass, redshift, BGG/satellite status, quartet luminosity, and σ_v (Table 4). It is clearest against Control_{4B} and RG₄ and weakest against the projected cores Control_{4C}, without implying that the three estimates differ significantly from one another; omitting the noisy σ_v covariate leaves the interpretation unchanged (Appendix H). It is carried by the satellites: the adjusted BGG odds ratios are 0.78, 0.87, and 1.87, with intervals that include unity. A quartet-level version of the comparison, which models the number of E-class satellites of every quartet, gives the same ordering (Appendix G).

In the pooled model over the deduplicated control catalogues (Appendix G), the satellite E-class odds ratio is 2.08 (95% CI [1.51, 2.88]) ($p < 10^{-4}$), whereas the BGG estimate is 0.78 (95% CI [0.39, 1.53]) ($p = 0.464$); the pooled all-galaxy odds ratio is 1.77 ($p = 1.4 \times 10^{-4}$).

3.2. Position within the group

Compact-group satellites sit much closer to their BGG in projection than control satellites (Fig. 3, lower panel). Their median projected distance is 123 kpc and none lies beyond 319 kpc, whereas the median distances of Control_{4B}, Control_{4C}, and RG₄ satellites are 381, 226, and 338 kpc, and only 42, 65, and 46% of them lie within 319 kpc. In every sample the E-class fraction declines with projected distance from the BGG (Fig. 3, upper panel), as in the morphology–density and morphology–radius relations of groups and clusters (Postman & Geller 1984; Whitmore et al. 1993), and at a given distance the CG₄ satellites lie at or above the controls. Adding $\log R_{\text{BGG}}$ to the satellite models lowers the CG₄ odds ratios from 2.63, 1.93, and 3.03 to 1.80, 1.39, and 1.95 (Table 4). Expressed as fractions (Table 5), galaxies with the stellar masses, redshifts, and quartet properties of the CG₄ satellites would have E-class fractions of 0.34, 0.41, and 0.32 if they followed the three control samples, and 0.43, 0.49, and 0.42 if they also had the projected distances of the CG₄ satellites, against the observed 0.56. The CG₄-minus-control difference therefore falls from 0.22, 0.15, and 0.24 to 0.13, 0.07, and 0.14 at common projected distance (Table 5). Scaling the distance by r_{200} of the host halo, using a spline in $\log R_{\text{BGG}}$, or restricting the controls to the distance range of the CG₄ satellites gives the same picture (Appendix H). Because the projected distance of a compact-group satellite is partly set by the compactness criterion itself, the contrast at common distance is a different quantity from the total contrast, not a corrected version of it.

Within their own hosts the comparison can be made without any difference between host groups. For the 19 embedded and 37 predominant systems, whose quartets live inside richer Lim groups (54 hosts, 958 member galaxies, 224 of them in a CG₄), a conditional logistic regression stratified by host compares CG₄ members with the other members of the same host at fixed stellar mass and host lu-

Table 2. sSFR-class counts

Sample	Quenched	Star-forming	No sSFR
CG ₄	146 (63%)	84 (37%)	18 (7%)
BGG	45 (80%)	11 (20%)	6 (10%)
Satellites	101 (58%)	73 (42%)	12 (6%)
Control _{4B}	1593 (61%)	1019 (39%)	180 (6%)
BGG	533 (83%)	107 (17%)	58 (8%)
Satellites	1060 (54%)	912 (46%)	122 (6%)
Control _{4C}	1560 (60%)	1061 (40%)	191 (7%)
BGG	537 (83%)	107 (17%)	59 (8%)
Satellites	1023 (52%)	954 (48%)	132 (6%)
RG ₄	101 (46%)	118 (54%)	5 (2%)
BGG	38 (72%)	15 (28%)	3 (5%)
Satellites	63 (38%)	103 (62%)	2 (1%)
SDSS reference	7705 (15%)	44192 (85%)	634 (1%)

Notes. Quenched and star-forming percentages use classified galaxies as the denominator; No-sSFR percentages use all galaxies in the row. The SDSS non-AGN sample is a deeper orientation reference, not a control sample and enters no test.

Table 3. Morphology-class counts

Sample	GZ1 E	GZ1 S	Unc.	No GZ
CG ₄	124 (50%)	86 (35%)	7 (3%)	31 (12%)
Control _{4B}	1106 (40%)	1266 (45%)	116 (4%)	304 (11%)
Control _{4C}	1118 (40%)	1229 (44%)	144 (5%)	321 (11%)
RG ₄	60 (27%)	129 (58%)	15 (7%)	20 (9%)
SDSS ref.	10091 (19%)	34715 (66%)	7725 (15%)	0 (0%)

Notes. Percentages use each sample’s final row count. Inferential E/S comparisons condition on the usable E/S subset. “No GZ” denotes missing vote fractions and is distinct from finite-vote uncertain classifications. The SDSS reference is for orientation only.

Table 4. Adjusted CG₄ odds ratios against each control

Outcome	Galaxies and added terms	Control _{4B}	Control _{4C}	RG ₄
E versus S	All galaxies	2.14 [1.58, 2.90]	1.67 [1.25, 2.24]	2.81 [1.79, 4.38]
	BGGs	0.78 [0.40, 1.54]	0.87 [0.44, 1.75]	1.87 [0.63, 5.52]
	Satellites	2.63 [1.88, 3.68]	1.93 [1.40, 2.66]	3.03 [1.72, 5.34]
	Satellites, + log R_{BGG}	1.80 [1.28, 2.55]	1.39 [1.01, 1.92]	1.95 [1.08, 3.53]
	Satellites, +sSFR class	1.94 [1.30, 2.90]	1.68 [1.14, 2.47]	2.41 [1.26, 4.63]
Quenched versus star-forming	All galaxies	1.80 [1.26, 2.58]	1.25 [0.91, 1.71]	1.40 [0.80, 2.47]
	Satellites	2.16 [1.51, 3.09]	1.38 [1.00, 1.90]	1.85 [0.99, 3.46]
	Satellites, +GZ1 class	1.85 [1.13, 3.02]	1.29 [0.83, 2.01]	1.39 [0.65, 2.97]
	Satellites, +GZ1 class + log R_{BGG}	1.49 [0.88, 2.53]	1.07 [0.67, 1.70]	1.11 [0.49, 2.51]

Notes. Odds ratios of the CG₄ indicator with 95% confidence intervals, from separate binomial models of CG₄ and one control. All models include stellar mass, redshift, quartet luminosity, and σ_v , and the all-galaxy models a BGG/satellite indicator; R_{BGG} is the projected distance to the BGG. Models with the other classification as a covariate use the galaxies classified in both. Standard errors are clustered by physical Lim group.

minosity rank. Without a distance term, CG₄ members are more often in the E class than their co-members (odds ratio 1.90 [1.23, 2.95]). The two populations, however, occupy different parts of their hosts: the median projected distance from the host BGG is 142 kpc for member satellites and 634 kpc for the other satellites, and only 17% of the latter lie within the largest member distance of their own host. At a given distance the E-class fractions of members and co-members are similar (satellite odds ratio 0.93 [0.55, 1.57] with log R as covariate; Appendix H), but with so little overlap in distance the within-host comparison cannot separate a compact-group effect from the radial trend of the host.

A local-density proxy tells a similar story. For each of the 6076 catalogue rows we compute a pairwise tidal index $T_i = \sum_j M_{\star,j}/R_{ij}^3$ over the selected quartet co-members (R_{ij} in kpc; co-members without a stellar mass contribute zero, so T_i is a lower bound in those groups) and refit the GZ1 E-class model, adjusted for stellar mass and satellite status and clustered by physical group, with and without the index on the three control catalogues pooled *without* objid deduplication (unlike the pooled models of Appendix G). The CG₄ odds ratio falls from 1.95 ($p < 10^{-4}$) to 1.39 ($p = 0.031$) when the index is added, the index itself having an odds ratio of 1.42 per 1.07-dex standard deviation of log T_i ($p < 10^{-4}$). Because the index is built from the same

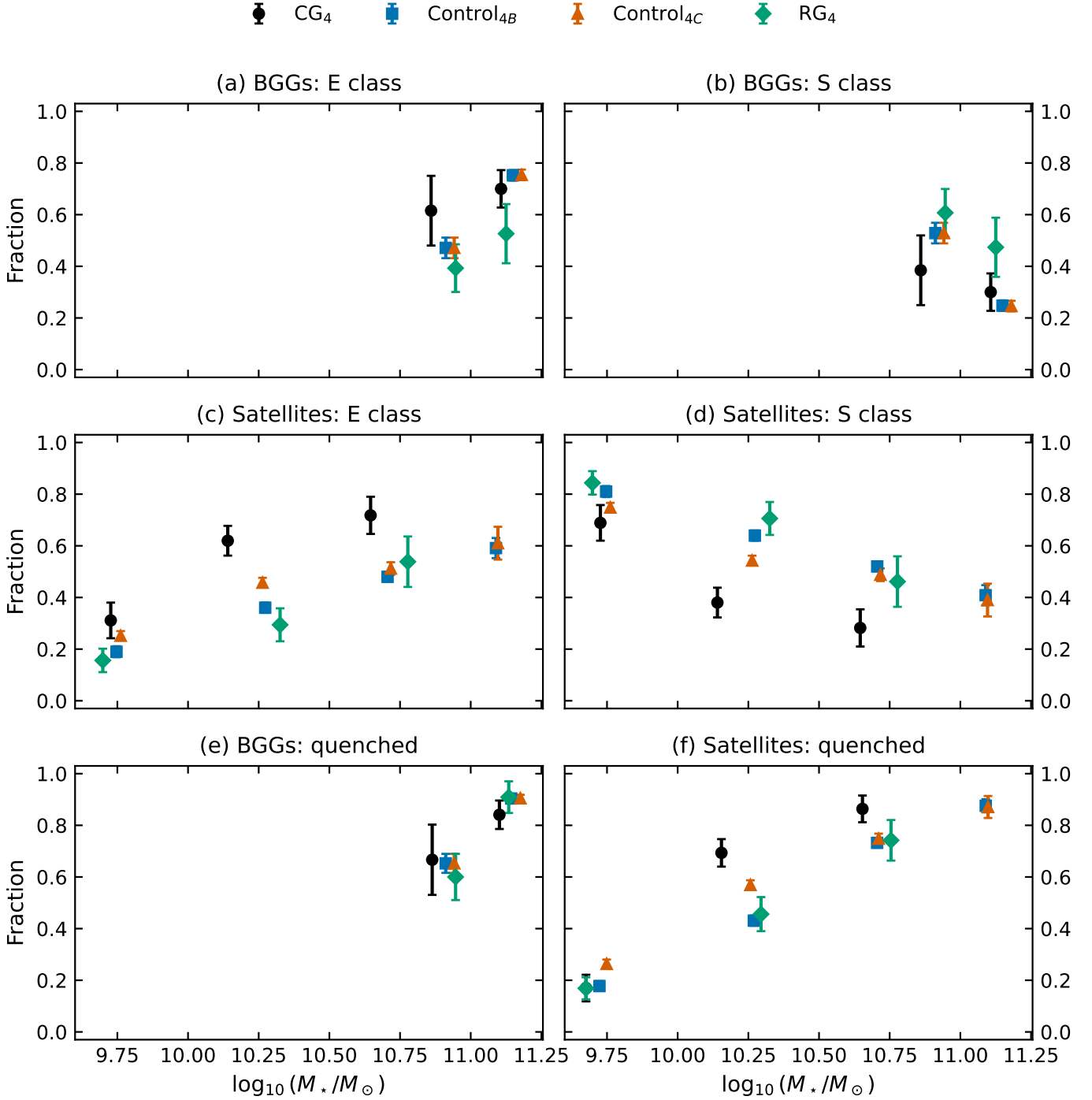


Fig. 2. Morphology and quenched fractions versus stellar mass. Rows show BGG E/S fractions, satellite E/S fractions, and quenched BGG/satellite fractions. All six panels use edges 7.0, 10.0, 10.5, 11.0, 12.5; points are placed at the median mass in each bin. Error bars are binomial 68% intervals, with Wilson limits for empty or full bins; 68% intervals from a bootstrap that resamples groups have almost the same widths (median ratio 1.00). Bins with fewer than five galaxies or three physical groups are omitted.

projected separations that define a compact group, holding it fixed partly holds the selection fixed, so the attenuation does not show that local density produces the excess. The residual CG₄ term is also unstable on common support: the CG₄ and control log T_i distributions overlap with coefficient 0.54 but their medians differ by 1.1 dex, and the residual odds ratio becomes 1.40 with a spline in log T_i , 1.27 after deduplicating the control pool ($p = 0.12$), and 1.22 on the common-support window ($p = 0.20$). The index is a proxy

for immediate neighbours in projection, not a measurement of tidal history; the restriction to quartet co-members is innocuous (Appendix G.4).

3.3. Star formation at fixed morphology

The quenched-fraction contrast is weaker and more control-dependent than the morphological one (Table 4): the satel-

Table 5. Satellite E-class fractions for the covariates of the CG₄ satellites

	Control _{4B}	Control _{4C}	RG ₄
Control satellites	0.40	0.41	0.27
Control relation	0.34	0.41	0.32
	[0.31, 0.37]	[0.37, 0.44]	[0.24, 0.42]
with CG ₄ R_{BGG}	0.43	0.49	0.42
	[0.38, 0.47]	[0.44, 0.53]	[0.32, 0.53]
CG ₄ minus control	0.22	0.15	0.24
	[0.14, 0.30]	[0.08, 0.23]	[0.12, 0.35]
at common R_{BGG}	0.13	0.07	0.14
	[0.05, 0.21]	[0.00, 0.15]	[0.02, 0.26]

Notes. The observed fraction $N_E/(N_E + N_S)$ of the 157 classified CG₄ satellites is 0.56. The model rows average, over the CG₄ satellites, the E-class probability given by the satellite model of Table 4 without and with $\log R_{\text{BGG}}$, with the CG₄ indicator set to zero. Brackets give 95% intervals from 2000 bootstrap draws that resample physical groups.

lite odds ratios are 2.16, 1.38, and 1.85 against Control_{4B}, Control_{4C}, and RG₄, and assigning all missing-sSFR galaxies to either extreme class changes them little (Appendix B).

Part of this contrast is associated with the different morphology mix, but stellar mass has to be held fixed at the same time: within each GZ1 class the CG₄ satellites are less massive than the Control_{4B} satellites (median $\log(M_*/M_\odot)$ of E-class satellites 10.34 against 10.62), and the quenched fraction rises steeply with mass. At fixed GZ1 class and stellar mass, CG₄ satellites are more often quenched than Control_{4B} satellites (odds ratio 1.85 [1.13, 3.02]; a quenched fraction of 0.61 against 0.53 expected from the Control_{4B} relation), whereas the odds ratios against Control_{4C} and RG₄, 1.29 [0.83, 2.01] and 1.39 [0.65, 2.97], are consistent with no difference. With the projected BGG distance also held fixed, the Control_{4B} value becomes 1.49 [0.88, 2.53]. Conversely, the E-class excess persists at fixed sSFR class against all three controls (Table 4). Such single-epoch associations cannot tell whether the morphological change preceded the quenching or accompanied it. Alternative sSFR classifications and a decomposition by morphology and mass bins do not change these conclusions (Appendix B).

No robust CG₄ displacement of the star-forming main sequence is established: the unmatched median offsets are comparison-dependent, while the matched star-forming pairs give a small mean difference whose interval includes zero. The cumulative distributions and full numerical comparison are retained in the online supplement. The quenched-sequence residuals are also shown there, but are weakly informative because MPA-JHU sSFRs of quenched galaxies are largely D_n4000 -calibrated.

3.4. Other galaxy properties

None of the other galaxy properties shows an offset as robust as the morphology excess (Fig. 4). Strong H α emission, an equivalent width $W(\text{H}\alpha) \geq 3 \text{ \AA}$ with the line detected at $S/N \geq 3$, is rarer in CG₄: 34% of the members with line measurements against 41%, 43%, and 56% in Control_{4B}, Control_{4C}, and RG₄ (Fisher $p = 0.05$, $p = 0.007$, $p < 10^{-4}$). The deficit persists at fixed stellar mass and GZ1

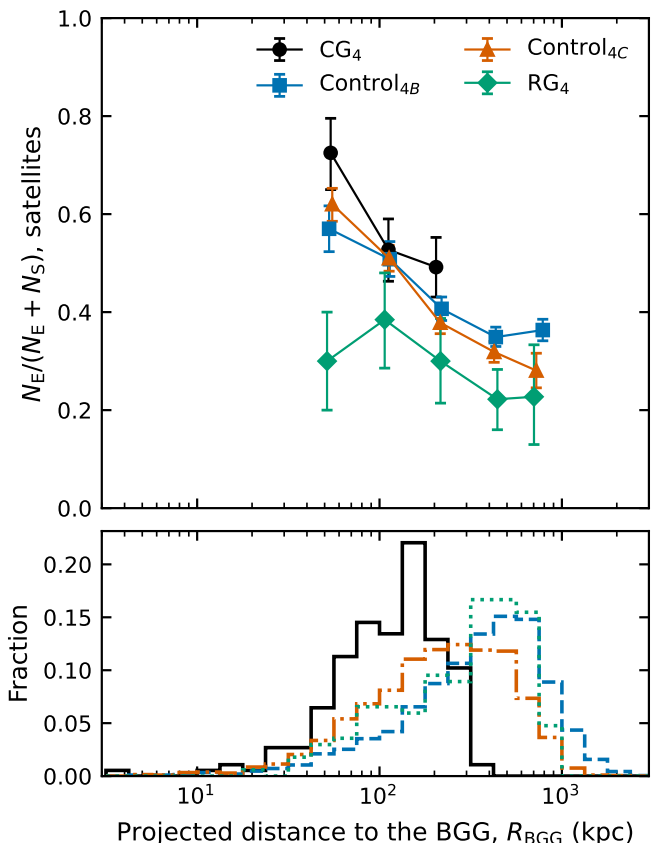


Fig. 3. Satellites and their projected distance from the BGG. Upper panel: E-class fraction $N_E/(N_E + N_S)$ in bins of projected distance (edges 0, 75, 150, 300, 600, 3000 kpc), plotted at the median distance in each bin, with 68% intervals from a bootstrap that resamples groups; bins with fewer than five galaxies or three groups are omitted. Lower panel: distribution of the projected distances of the satellites of each sample.

class, with odds ratios of 0.55, 0.65, and 0.54. Splitting the strong emitters at $\log([\text{N II}]/\text{H}\alpha) = -0.4$, the separation between star-forming and AGN-like emission in the WHAN diagram of Cid Fernandes et al. (2011), shows where it comes from. At fixed mass and class, star-forming-like emission is not rarer in CG₄ than in the controls (odds ratios 1.25 [0.75, 2.07], 1.57 [0.96, 2.56], and 1.59 [0.78, 3.24]; Fig. 4a), and its H α equivalent widths and fibre luminosities differ from those of the controls by at most 0.07 dex at fixed mass, whereas strong emission with high $[\text{N II}]/\text{H}\alpha$ is rarer (0.38 [0.23, 0.63], 0.40 [0.25, 0.66], and 0.37 [0.19, 0.72]; Fig. 4b), weak-line galaxies taking its place. This high-ratio class mixes LINER- and Seyfert-like, composite, and metal-rich star-forming spectra: among the strong emitters with all four BPT lines at $S/N \geq 3$, 79% of the CG₄ galaxies lie in the star-forming region of the diagram, against 60%, 67%, and 69% in the controls. It is therefore not an AGN census, and because we examined this decomposition only after finding the deficit, we regard it as exploratory. The D_n4000 index follows stellar mass in every sample; the CG₄ medians sit above the controls only in the 10.0–10.5 bin ($D_n4000 = 1.77$ against 1.59, 1.68, and 1.56), where the CG₄ E-class fraction also peaks relative to the controls, and coincide with them elsewhere; a post-starburst classification from D_n4000 and $\text{H}\delta_A$ is deferred to a follow-up paper.

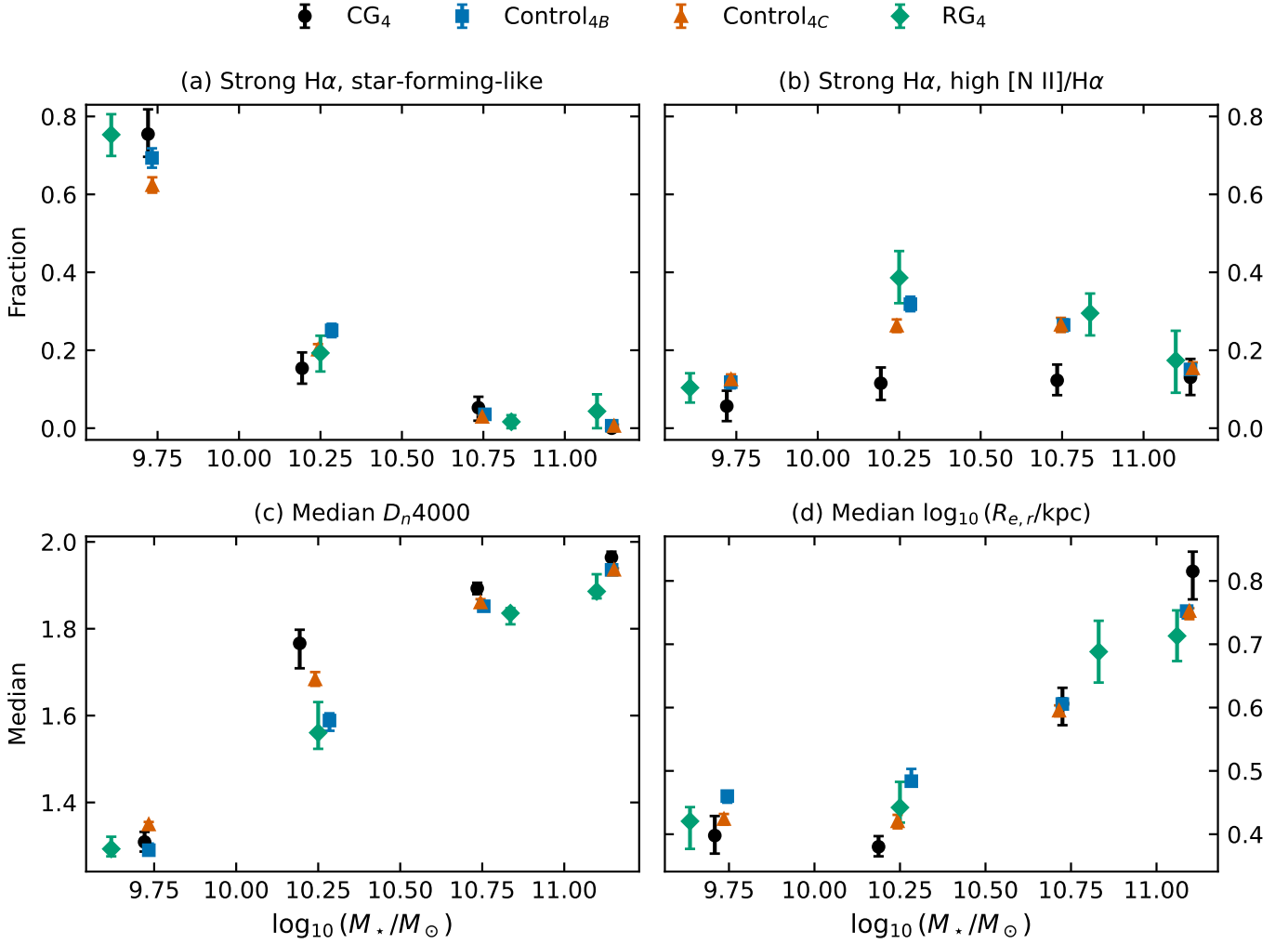


Fig. 4. Other galaxy properties versus stellar mass: fractions of galaxies with strong H α emission and (a) $\log([\text{N II}]/\text{H}\alpha) < -0.4$ (star-forming-like) or (b) $\log([\text{N II}]/\text{H}\alpha) \geq -0.4$, whose sum is the strong-emission fraction; (c) median D_n4000 ; (d) median seeing-corrected effective radius $\log_{10}(R_{e,r}/\text{kpc})$. Bins follow Fig. 2; error bars are 68% intervals from a bootstrap that resamples groups. These comparisons are descriptive.

CG₄ galaxies are not smaller than ordinary-group galaxies of the same stellar mass. With the seeing-corrected Simard radii, the satellite size offset at fixed stellar mass, redshift, and group-scale covariates is -0.010 dex (-2% ; 95% CI $[-0.039, 0.019]$; $p = 0.51$; 2575 satellites in 754 physical groups), and -0.006 dex for all galaxies ($p = 0.65$); the per-control estimates are -0.031 dex against Control_{4B}, -0.001 dex against Control_{4C}, -0.021 dex against RG₄ (Appendix I; see the online supplement). The one-to-one matched difference is -0.019 dex for the 166 pairs with Simard sizes (95% CI $[-0.064, 0.026]$; $p = 0.40$), and -0.033 dex with the uncorrected Petrosian radii ($p = 0.03$). The Simard measure resolves 77% of the CG₄ galaxies against 71% of the pooled controls, so the Simard coefficients carry a differential-completeness caveat that the Petrosian rerun does not share (Petrosian satellite coefficient -0.019 dex, $p = 0.11$). Colours, magnitude gaps, and projected phase-space trends are likewise control-dependent or weak; they are reported in the online supplement.

3.5. Robustness

The morphology excess does not depend on the vote threshold, on the classifier, on the closest pairs, or on how the comparison is organised (Table 6). Excluding every galaxy with another galaxy of the Lim catalogue within 55 arcsec, the SDSS fibre-collision scale, removes 27% of the CG₄ members against 10–16% of the controls, leaves the E-class fraction $N_E/(N_E + N_S)$ at 0.58 in CG₄ against 0.45, 0.46, and 0.31 in the controls, and moves the adjusted odds ratios from 2.14 to 2.09 (Control_{4B}), 1.67 to 1.69 (Control_{4C}), and 2.81 to 2.78 (RG₄). Across classification thresholds of 0.4–0.8 the pooled adjusted odds ratio stays between 1.73 and 2.19, the continuous debiased E vote fraction p_E itself is offset by 0.089 ($p < 10^{-4}$), and the Galaxy-Zoo-independent Sérsic index ($n_g > 2.5$; 75% coverage) gives an odds ratio of 1.60 ($p = 0.002$). With the DS18 $T\text{Type} \leq 0$ classification as the outcome, the per-control CG₄ odds ratios are 2.11, 1.49, and 2.22 (all $p \leq 0.017$; $n = 1822, 1913, 324$). The quartet-level models, the one-to-one matches, and the crowding, sample-definition, and σ_v reruns are detailed in Appendices G and H.

Table 6. Morphology-contrast robustness

Test	Result
No neighbour within 55 arc-sec	OR 2.09 / 1.69 / 2.78
DS18 T Type ≤ 0 outcome	OR 2.11 / 1.49 / 2.22
Vote thresholds 0.4–0.8 (pooled)	OR 1.73–2.19
Continuous p_E (pooled)	0.089 [0.047, 0.131]
Sérsic $n_g > 2.5$ (pooled)	OR 1.60; $p = 0.002$
Without σ_v	OR 2.22 / 1.68 / 2.76
Quartet level, all quartets	OR 2.14 / 1.80 / 2.72
Matched quartets	Δ 0.16 / 0.11 / 0.20

Notes. Adjusted CG_4 odds ratios for E-class membership; triplets refer to $Control_{4B} / Control_{4C} / RG_4$, pooled rows to the deduplicated control pool. The continuous row is the adjusted offset in debiased E vote fraction, the quartet rows model the number of E-class satellites per quartet, and Δ is the matched difference in satellite E-class fraction (Appendix G).

4. Interpretation and comparison with previous work

4.1. An early-type-like satellite excess linked to dense group cores

Compact groups of four galaxies host more early-type-like satellites than any of the three ordinary-group comparison samples, while their BGGs resemble those of the controls. The three adjusted contrasts are ordered in the same sense as the characteristic compactness of the controls, with the smallest contrast against the projected cores $Control_{4C}$; this is a descriptive ordering, not a formal between-control trend (Sect. 3.1). Much of the contrast is positional. Compact-group satellites sit where early-type-like galaxies are common in every group, close to the BGG, and the decline of the E-class fraction with projected distance that we find in all four samples is the morphology–density relation as seen within groups (Postman & Geller 1984). At common projected distance a difference remains against all three controls (Sect. 3.2), so compact-group satellites are more early-type-like than control satellites at the same distance from their BGG. Within their own hosts, compact-group members resemble their co-members at the same distance, but that comparison has little leverage because the two populations hardly overlap in radius. The two results would be reconciled if the inner regions of the groups that host compact groups were more evolved than those of typical groups, rather than the compact-group members differing from their immediate neighbours, but the data cannot separate these possibilities. The within-host comparison extends the surrounding-host analysis of Montaguth et al. (2026b) by asking whether morphology differs beyond shared host and central location; samples, morphology definitions, and structural measurements differ, so the results are complementary. Present-day stellar mass may itself encode part of a galaxy’s environmental history, so fixed-mass estimates describe the remaining association rather than the total accumulated transformation.

The local projected-density comparison of Sect. 3.2 is consistent with this picture without proving it: the tidal-index term is strongly associated with E-class morphology

and attenuates the CG_4 coefficient (from 1.95 to 1.39), as expected if compact groups select galaxies in locally dense projected configurations where repeated close companions, rather than the catalogue label, are linked to the early-type-like population. Because the index is entangled with the selection and its residual term is not stable on common support, a projected snapshot cannot show that local density drives the morphological difference. This reading is consistent with compact groups tracing the high-density, high-interaction tail of group environments—some genuinely compact, others dense cores or substructures inside larger haloes—and with the stronger structural signatures reported for non-isolated compact groups in recent S-PLUS work (Montaguth et al. 2023, 2025, 2026a,b). The 6 isolated compact groups have few E-class satellites (Appendix F.1), but they are too few to test a contribution from the larger-scale host, and adding the host halo mass leaves the adjusted contrasts essentially unchanged (Appendix G.4).

Stellar-population diagnostics could in principle separate an accumulated enrichment history from a shorter star-formation timescale, but we defer a stellar-metallicity analysis for two reasons. The quenching excess at fixed morphology is modest and depends on the control sample (Sect. 3.3), so a strangulation-type metallicity test of the kind proposed by Peng et al. (2015) would have little leverage here, and the single-fibre SDSS spectra sample only the central ~ 2 kpc of the galaxies at the sample redshifts, where the gradients of both metallicity and age are steepest.

4.2. $GZ1$ E-class morphologies and transition galaxies

Our E class is a smooth early-type-like proxy with a substantial S0 content (Sect. 2.2). Galaxy Zoo 1 cannot determine whether the excess comprises S0s, ellipticals, or both; we reserve that question for a dedicated, pre-registered analysis of the DS18 classes by environment. This caveat matters when comparing with the S-PLUS studies of compact-group transition galaxies and R_e –Sérsic-index trends (Montaguth et al. 2023, 2025, 2026a,b): their structural measurements and transition populations probe pathways, whereas our three control samples test which apparent CG_4 signatures survive across genuine four-member, luminosity-selected, and BGG-centred projected-core quartets.

4.3. Why the star-formation and size signals are weaker

Beyond morphology, the evidence for a compact-group effect on star formation is weaker. At fixed morphology and stellar mass the quenched fraction of CG_4 satellites exceeds that of $Control_{4B}$ satellites, but it is not detectably higher than in the projected cores $Control_{4C}$ or in RG_4 , and the $Control_{4B}$ difference weakens at fixed projected distance; the quenched odds ratios of Table 4 therefore combine the different morphology mixes with, at most, a modest and control-dependent excess. The $H\alpha$ deficit points elsewhere: star-forming-like emission is not rarer in CG_4 than in the controls at fixed mass and morphology, and the missing strong emitters are those with high $[NII]/H\alpha$. In the controls many such spectra are composites or LINER- and Seyfert-like systems, in which star formation, if present, coexists with other sources of ionisation;

their scarcity in CG₄ could reflect fewer galaxies caught in that mixed state, weaker nuclear or extended low-ionisation emission, or aperture differences, and single-fibre line ratios cannot decide among these. The spectroscopic measurements are also affected by crowding. CG₄ members have another catalogued galaxy within the 55-arcsec fibre-collision scale about twice as often as control members (27 against 14%), such galaxies lack an sSFR about 10 times more often than the others, and companions that never received a spectrum are absent from the catalogues altogether. Assigning the missing galaxies to either extreme class leaves the adjusted contrasts qualitatively unchanged, but short-lived starburst or post-starburst phases in the closest pairs could be under-represented, and the MPA-JHU aperture corrections (Brinchmann et al. 2004; Salim et al. 2007) may be less secure for interacting or blended members. We therefore read the sSFR and main-sequence results as conservative, not as proof that compact groups leave star formation unaffected.

The size result carries a parallel message. Coenda et al. (2012) found compact-group galaxies to be smaller at fixed luminosity than loose-group and field samples, especially early types, using SDSS Petrosian half-light radii; against three regular-group controls, with seeing-corrected model sizes, cluster-robust adjusted models, one-to-one matching, and crowding tests, we find no size offset at fixed stellar mass (satellite coefficient -0.010 dex, bounding any mean CG₄ size deficit to roughly 0.04 dex; matched Simard difference -0.019 dex). This is compatible with a compact-group versus field offset if ordinary-group cores have already undergone comparable structural processing. A negative fixed-mass size offset could reflect tidal truncation but also higher bulge dominance or residual blending in crowded fields; the morphology-stratified fits, the Petrosian cross-check, and the crowding exclusions of Appendix I bound both readings.

5. Conclusion

We have tested whether galaxies in CG₄ compact groups remain special after removing split systems and comparing them with RG₄, Control_{4B}, and Control_{4C}. The conclusions apply to this luminous, volume-limited, exactly four-member non-split HMCg sample rather than to all compact groups.

The main result is an early-type-like satellite excess. CG₄ satellites are more often in the GZ1 E class than the satellites of all three regular-group comparisons after accounting for galaxy and group properties; the difference is strongest against Control_{4B} and RG₄, weakest against the projected cores Control_{4C}, and absent for BGGs. Much of it is associated with the central positions of the compact-group satellites: at their projected distances from the BGG, the control samples predict E-class fractions of 0.43, 0.49, and 0.42, against the observed 0.56. Within their hosts, compact-group members resemble co-members at the same distance, but the two populations overlap too little in radius for that comparison to be decisive. A projected tidal-density proxy likewise attenuates the CG₄ coefficient, but it is entangled with the compact-group selection.

Evidence for additional quenching is weaker and depends on the comparison: at fixed morphology and stellar mass, CG₄ satellites are more often quenched than Control_{4B} satellites but not detectably more than Control_{4C}

or RG₄ satellites, and star-forming CG₄ galaxies show no robust main-sequence offset. The strong-H α deficit is a deficit of strong emission with high [N II]/H α rather than of star-forming-like emission, an exploratory result whose physical origin remains open. At fixed stellar mass the seeing-corrected satellite size estimate is -0.010 dex with an interval including zero, so we do not claim a compact-group size effect. Colours, magnitude gaps, D_n4000 , and projected phase-space trends identify no simple alternative driver.

We therefore interpret CG₄ systems as hosting satellites shifted towards early-type-like morphologies, in part as expected for galaxies near the centres of groups but with an excess that remains at fixed projected position, consistent with structural transformation accumulated in dense group cores and with preprocessing. The data do not establish whether this transformation preceded the quenching or accompanied it. Explicit E–S0–spiral classifications, bulge–disc structure, and stellar-population diagnostics will be needed to determine whether the satellite excess reflects disc fading, tidal heating, or merger-driven transformation.

Data availability

The input data are public SDSS, MPA-JHU, Galaxy Zoo, Simard et al. (2011), Domínguez Sánchez et al. (2018), group, and compact-group catalogues cited above. Derived galaxy and group tables will be made available through CDS at publication. The analysis and paper-generation code is available at https://github.com/MatthieuTricottet/CG_Gals; `reproduce.py` regenerates every number, figure, and table of this paper from the tracked inputs, including the cached SDSS retrievals (`src/data_loader.py`, `src/ssfr_quality_audit.py`, `src/spectral_indices.py`, `src/emission_lines.py`). The article release, including the online supplement, will be archived with a DOI at acceptance.

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Appendix A: SDSS data query

For the SDSS reference sample, we extract `sfr_tot_p50`, `specsfr_tot_p50`, and `lgm_tot_p50` from `galSpecExtra`, `h_alpha_eqw` and the `[N II]`, `H $\alpha$` , `[O III]`, and `H $\beta$` fluxes from `galSpecLine`, and `p_el_debiased` and `p_cs_debiased` from `zooSpec`. In that query, these spectroscopic tables are joined to `SpecObj` by `specObjID`, while the photometric object is linked through `SpecObj.bestObjID=PhotoObj.objID`. The full result is split into the AGN-inclusive table and the non-AGN reference used to calibrate the sSFR classes. The group samples instead inherit their DR12 `sfr_tot_p50`, `lgm_tot_p50`, and stored `specObjID` from the parent catalogues; the Galaxy Zoo vote fractions are added to those rows by photometric `objid`, and the `galSpecIndx` indices `d4000_n` and `lick_hd_a` (with errors; sentinel values and non-positive errors treated as missing) and the `galSpecLine` measurements by the stored DR12 `specObjID`. Their final `log sSFR` is the stored difference `sfr_tot_p50-lgm_tot_p50`.

We audited all 6076 final sample rows against `SpecObjAll` in DR12, retaining the stored `specObjID` even when the same photometric `objid` has several spectra (looking these DR12 identifiers up in DR16 fails for reprocessed BOSS observations, whose identifier encodes the reduction version). Of the 394 rows without a valid sSFR, 332 are BOSS-instrument spectra (`survey = boss`, `run2d = v5_7_0`) without any `galSpecExtra` or `galSpecInfo` row, because the MPA-JHU tables cover the legacy `run2d = 26` reductions only; 60 are legacy spectra with a `galSpecExtra` row, a valid stellar mass, `reliable = 1`, and `zWarning = 0`, whose `sfr_tot_p50` is the `-9999` sentinel; and 2 (both in `Control4B`) have no stored spectrum identifier. Per sample the BOSS/sentinel counts are 14/4 in `CG4`, 149/29 in `Control4B`, 164/27 in `Control4C`, and 5/0 in `RG4`. The missing rows are not low-signal-to-noise spectra: their median `snMedian_r` is 45 against 39 for the classified rows (rank-sum $p < 10^{-4}$ over unique spectra), and the same holds within each sample. The missing class is therefore a coverage gap of the MPA-JHU value-added catalogues plus a small number of failed SFR fits, but it is not independent of environment. Of the missing rows, 65% have another galaxy of the [Lim et al. \(2017\)](#) catalogue within 55 arcsec, against 12% of the classified ones. In a logistic model of missingness on the deduplicated rows with the r -band absolute magnitude (available for every row, unlike the MPA-JHU stellar mass, which is missing for 84% of the missing rows), redshift, the BGG/satellite indicator, and `CG4` membership, a close neighbour raises the odds of a missing sSFR by a factor of 14 (95% CI [10, 20]) and brighter galaxies are more often missing, whereas BGG status (odds ratio 1.37 for satellites) and `CG4` membership (0.72) add little. The same galaxies usually lack Galaxy Zoo votes as well (90% of the missing rows), because the votes come from the `zooSpec` table, which lists galaxies with legacy spectra. The extreme-assignment bounds of Appendix B therefore matter more for crowded systems than the overall missing fractions suggest.

The emission-line fluxes and equivalent widths of `H $\alpha$` , `[N II] $\lambda$ 6584`, `H $\beta$` , and `[O III] $\lambda$ 5007`, with their uncertainties, are the `galSpecLine` values of the stored spectrum of each group galaxy, retrieved by `specObjID`; for every galaxy with line data this is the spectrum that also provides its sSFR.

Appendix B: sSFR classification details

We identify BPT-classifiable galaxies by requiring measured values of both emission-line ratios

$$\log_{10}([\text{N II}]\lambda 6584/\text{H}\alpha) \quad \text{and} \quad \log_{10}([\text{O III}]\lambda 5007/\text{H}\beta).$$

Among these galaxies, we flag AGN-like systems if they satisfy either

$$\log_{10} \frac{[\text{N II}]}{\text{H}\alpha} > 0,$$

or if they lie above the empirical demarcation of [Kauffmann et al. \(2003a\)](#):

$$\log_{10} \frac{[\text{O III}]}{\text{H}\beta} > \frac{0.61}{\log_{10}([\text{N II}]/\text{H}\alpha) - 0.05} + 1.3,$$

in which case they are excluded from the calibration sample. This calibration flag is deliberately more inclusive than the three-class BPT scheme of Sect. 3.4, in which composite galaxies lie between the [Kauffmann et al. \(2003a\)](#) and [Kewley et al. \(2001\)](#) demarcations and only galaxies above the [Kewley et al. \(2001\)](#) line are labelled AGN-like, because its sole purpose is to keep possible AGN contamination out of the sSFR calibration. Galaxies with missing diagnostic line ratios are BPT-unclassified, not proven non-AGN; they are retained for the sSFR status assignment when valid SDSS stellar mass and sSFR estimates are available. Reference-sample rows with unavailable `specsfr_tot_p50`, and group-sample rows with unavailable `sfr_tot_p50` or `lgm_tot_p50`, including the `-9999` sentinel, have no valid sSFR estimate; they are treated as missing data throughout: excluded from the mixture fit where applicable, never assigned an sSFR class, omitted from every sSFR figure, and reported as counts only (Sect. 2.2). Galaxies with a measured sSFR are divided into quenched and star-forming classes along the valley between the two populations of the SDSS non-AGN reference in the $\log M_{\star}$ – $\log$ sSFR plane. The valley is located with a two-component bivariate Gaussian mixture ([McLachlan & Peel 2000](#)) fitted by minimising the Kullback–Leibler divergence between the 50×50 -bin histogram of the reference sample and the mixture density, with the mean $\log$ sSFR of the low-sSFR component constrained to lie between `-12` and `-10` and its covariance determinant not allowed to exceed that of the star-forming component. Without these constraints, a maximum-likelihood fit to the AGN-filtered reference places both components on the star-forming sequence (mean $\log$ sSFR of `-10.86` and `-9.84`) and gives no usable boundary. The boundary is the locus where the two weighted component densities are equal (Fig. B.1). The reference sample uses the MPA-JHU `specsfr_tot_p50`, whereas the stored group values are `sfr_tot_p50-lgm_tot_p50`, higher by a median 0.05 dex for the same spectra. Applying the boundary to the `specsfr_tot_p50` of the group spectra instead, classifying with an unconstrained maximum-likelihood mixture fitted to the AGN-inclusive reference, or cutting at $\log(\text{sSFR}/\text{yr}^{-1}) = -11$ changes the class of 1.0, 2.5, and 4.1% of the group galaxies. Across the four classifications, the satellite odds ratio for quenching at fixed GZ1 class (Sect. 3.3) ranges from 1.74 to 2.05 against `Control4B`, from 1.19 to 1.32 against `Control4C`, and from 1.33 to 1.43 against `RG4`.

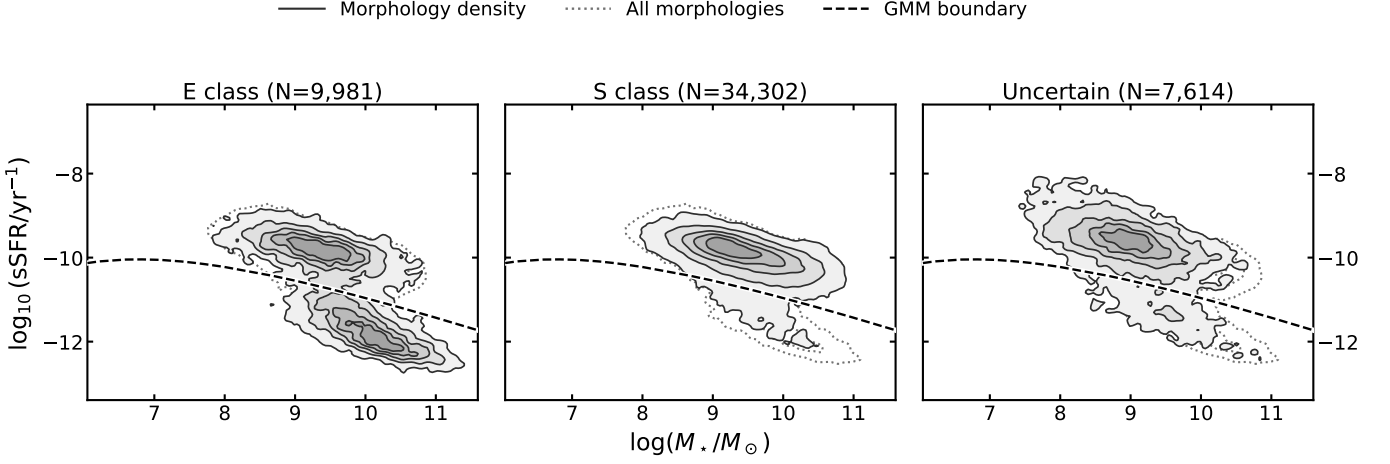


Fig. B.1. Galaxy Zoo morphology in sSFR–stellar-mass space for the SDSS non-AGN reference. Panels show the E, S, and uncertain classes. Solid class-specific contours enclose 20, 40, 60, 80, and 95% of the density after display-only Gaussian smoothing with $\sigma = 1.25$ histogram bins; grey dotted contours show the smoothed full reference distribution, and the dashed curve is the adopted boundary between quenched and star-forming galaxies.

Table B.1. Missing-sSFR extreme-assignment bounds

Scope	Control	Missing (%)	Δf_Q^{obs}	min	max	
		CG ₄	ctrl			
All	Control _{4B}	7.3	6.4	0.025	−0.046	0.091
All	Control _{4C}	7.3	6.8	0.040	−0.034	0.107
All	RG ₄	7.3	2.2	0.174	0.115	0.210
Sat.	Control _{4B}	6.5	5.8	0.043	−0.021	0.101
Sat.	Control _{4C}	6.5	6.3	0.063	−0.005	0.122
Sat.	RG ₄	6.5	1.2	0.201	0.156	0.233

Notes. Missing columns give the percentage without sSFR on each side. The observed difference uses classified galaxies; min/max assign all missing galaxies on either side to quenched or star-forming, using all-galaxy denominators.

Extreme-assignment bounds for the missing class. Because the missing-sSFR class could in principle hide differential quenching, Table B.1 bounds every raw quenched-fraction contrast by the four extreme assignments in which *all* missing-sSFR galaxies on *each* side are set to quenched or to star-forming, with consistent all-galaxy denominators on both sides. The missing fractions are similar in CG₄ and in Control_{4B} and Control_{4C} (7% of CG₄ members versus 6% and 7%) and differ only against RG₄ (2%). The raw RG₄ contrast survives all four corners (all-member difference between 0.12 and 0.21 around the observed 0.17; satellites 0.16 to 0.23), whereas the small raw Control_{4B} and Control_{4C} differences can change sign under the doubly adversarial corner. Propagated to the adjusted satellite models of Table 4, the four corners give quenched odds ratios of 1.96–2.30 against Control_{4B}, 1.28–1.46 against Control_{4C}, and 1.74–1.96 against RG₄, and 1.58–1.98, 1.14–1.36, and 1.22–1.43 at fixed GZ1 class; the missing galaxies mostly lack a GZ1 class as well, so the latter change little.

Star formation at fixed morphology. Table B.2 gives the quenched fractions among sSFR-classified galaxies in each GZ1 class. Within a class they are similar across the samples (homogeneity χ^2 test on $P(Q|E)$, $p = 0.39$), but

Table B.2. Conditional quenched fractions

	CG ₄	Control _{4B}	Control _{4C}	RG ₄
GZ1 E	0.908 ^{+0.040} _{−0.065}	0.903 ^{+0.016} _{−0.019}	0.887 ^{+0.017} _{−0.020}	0.850 ^{+0.069} _{−0.111}
Spiral	0.326 ^{+0.105} _{−0.090}	0.357 ^{+0.027} _{−0.026}	0.328 ^{+0.027} _{−0.026}	0.295 ^{+0.084} _{−0.072}

Notes. For the uncertain-morphology class, $P(Q|U) = 0.429$, 0.552, 0.594, and 0.267 ($n = 7, 116, 143$, and 15 classified galaxies, same sample order).

these fractions do not hold stellar mass fixed, and the CG₄ galaxies of each class are less massive than those of Control_{4B} (Sect. 3.3). A Kitagawa decomposition of the satellite quenched-fraction difference into a composition term and a conditional term illustrates this. With cells defined by GZ1 class alone, the conditional terms are −0.02, 0.00, and 0.07 against Control_{4B}, Control_{4C}, and RG₄; with cells defined by GZ1 class and the terciles of satellite stellar mass (edges $\log(M_*/M_\odot) = 10.09$ and 10.55), they become 0.07 [0.00, 0.14], 0.01 [−0.05, 0.07], and 0.03 [−0.05, 0.13] (95% intervals from a bootstrap over groups). The tercile edges are arbitrary, so the decomposition only illustrates the continuous-mass models of Sect. 3.3.

Appendix C: Compact groups as projected cores of ordinary groups

The compact-group exclusion of Sect. 2.1 is not a technicality: for 61 of the 765 parent Lim groups, the would-be Control_{4C} quartet (BGG plus the three closest projected companions within 3 magnitudes) contains at least one CG₄ galaxy—233 CG₄ galaxies in total. Table C.1 breaks this overlap down by the Zheng & Shen (2021) class: 37 of the 37 predominant and 14 of the 19 embedded compact groups *are* the projected cores of their host groups, as Fig. 1a illustrates; the remaining 5 embedded systems sit in rich hosts whose BGG-centred core is made of other members (Fig. 1b). The Control_{4C} quartets of 3 of these hosts contain no compact-group galaxy and are retained in the con-

Table C.1. Compact-group overlap with would-be Control_{4C} quartets

CG ₄ class	CG ₄ groups	Lim groups	Galaxies
Embedded	14 / 19	14	52
Isolated	6 / 6	6	24
Predominant	37 / 37	37	145
Split	4 / 16	4	12

Notes. “CG₄ groups” counts compact groups with at least one member inside a parent-group quartet, over the total number of compact groups of that class (split groups included, as in the exclusion).

trol sample (the other 2 hosts also contain a core-forming compact group), whereas all 5 would-be Control_{4B} quartets contain the compact group’s brightest member and are excluded. The isolated entries are trivial self-overlap: for each of the 6 isolated systems, the overlapping Lim group is the compact group’s own host, whose BGG and three nearest projected companions are exactly the four CG₄ members. Had these groups not been excluded, the same galaxies would have appeared on both sides of the comparisons, biasing every contrast toward null.

Appendix D: Statistical conventions

Independence unit. The unit for every inferential contrast among the group catalogues is the *physical* Lim group: standard errors are clustered by Lim group (unifying the overlapping control definitions, with CG₄ systems assigned to their host Lim group, or to their own identifier when they have no Lim counterpart), and bootstrap intervals resample physical groups, separately within CG₄ and the control sample, or matched pairs (quartet matching) and treated compact groups (galaxy matching). Pooled models across the three controls are secondary summaries computed after deduplicating the overlapping control samples to one row per physical galaxy, with the kept label following the priority RG₄ > Control_{4B} > Control_{4C} (Appendix G). Purely descriptive reference-sample diagnostics report their own unit where they appear.

Tests and p-values. Descriptive comparisons use two-sided Fisher exact tests for 2×2 tables and rank-sum or bootstrap comparisons for distributions and median differences; Spearman and Pearson coefficients are diagnostics of monotonic or linear trends only. Paired sign-flip permutation p-values use add-one tail counts, $p = (k + 1)/(B + 1)$ with k the number of permutations at least as extreme as the observation; bootstrap sign-crossing p-values are $2 \min[(k_{\leq 0} + 1)/(B + 1), (k_{\geq 0} + 1)/(B + 1)]$, capped at unity, where $k_{\leq 0}$ and $k_{\geq 0}$ count resampled differences on either side of zero; displayed p-values are floored at 10^{-4} . Adjusted odds ratios come from binomial generalised linear models with cluster-robust covariance; continuous covariates are standardised, so coefficients of continuous terms are per standard deviation of the fitted frame. Multiplicity control is reserved for exploratory screens: Holm adjustment for the colour, measurement-availability, and magnitude-gap screens, and Benjamini–Hochberg false-discovery-rate control for the dominance, morphology–environment, and second-order screens.

Covariates. The adjustment set is fixed: stellar mass, redshift, a BGG/satellite indicator (the only rank information used), the total r -band luminosity of the quartet, and the quartet velocity dispersion σ_v . The velocity dispersion is the gapper estimate (Wainer & Thissen 1976) on the four selected members’ redshifts, converted as $c \sigma_z / (1 + \bar{z})$, for all samples, including Control_{4B} and Control_{4C} quartets embedded in richer hosts (this is how the spurious Lim 3688 quartets of Sect. 2.1 reach $\approx 2000 \text{ km s}^{-1}$). With four tracers the estimator is intrinsically noisy: in a Gaussian Monte Carlo of 10^5 quartets, $\sigma_{\text{gapper}}/\sigma_{\text{true}}$ has median 0.96 with a 16–84% range of 0.57–1.42. The group-level covariates are complete for every quartet, and stellar mass is available for every galaxy with a GZ1 or sSFR class; the pipeline stops rather than dropping a covariate that is less than 65% complete in a model frame (70% for matching variables). No group-mass covariate is used (Appendix G.4).

Overlap coefficient. Where two distributions are compared for common support (compactness in Sect. 2.1, $\log T_i$ in Sect. 3.2), the overlap coefficient is the integral of the minimum of the two densities, estimated as 60-bin histograms over the common range; it is 0 for disjoint and 1 for identical distributions.

Fractions at the CG₄ covariates. For a binomial model fitted to CG₄ and one control, the control fraction at the CG₄ covariates is the mean, over the CG₄ galaxies, of the predicted probability with the CG₄ indicator set to zero; with the indicator set to one, the same average reproduces the observed CG₄ fraction. Their difference is a difference of fractions at the covariate distribution of the CG₄ galaxies, which, unlike an odds ratio, can be compared between models with different covariates. Intervals come from 2000 bootstrap draws that resample physical groups separately within CG₄ and the control, refitting the model at each draw.

Appendix E: Main-sequence and quenched-sequence details

We model the star-forming main sequence as a function of stellar mass with polynomials of order 1–4 fitted to the star-forming galaxies of the SDSS non-AGN reference sample; the diagnostic is the residual RMS scatter, not a formal reduced χ^2 , because no per-galaxy uncertainty model is used. We adopt order 2, the lowest order with the smallest five-fold cross-validated RMS (0.305 dex, against 0.306 dex for order 1), and compute the residual of each star-forming galaxy with respect to it; a positive residual means a higher sSFR at fixed stellar mass than the reference relation. In Sect. 3.3, Δ is the CG₄-minus-control median residual with the central 68% group-bootstrap interval on the median difference and the two-sided sign-crossing probability (10 000 resamples). The quenched sequence shown in the online supplement is obtained with the same procedure on the 7705 quenched galaxies of the reference sample: order 4 has the smallest cross-validated RMS (0.331 dex), and the residuals are attached to the quenched galaxies of the four samples.

Within CG₄, the star-forming main-sequence offsets of the embedded, isolated, and predominant classes are statistically consistent with one another.

Appendix F: Secondary and environmental diagnostics

This appendix collects the secondary diagnostics of the compact-group classes and of the group environment. Magnitude-gap, BGG-distance, dominance, and colour diagnostics are provided in the online supplement; Paper I contains the fuller group-scale comparison.

F.1. Compact-group classes

The satellite E-class fractions of the two non-isolated classes are similar, $f_E = N_E/(N_E + N_S) = 0.638^{+0.122}_{-0.143}$ for the 19 embedded and $0.562^{+0.095}_{-0.100}$ for the 37 predominant systems (Wilson 95% intervals), with a predominant-versus-embedded odds ratio at fixed stellar mass of 0.97 ($p = 0.94$), although their host haloes differ (median $\log M_{200c} = 13.83$ and 13.13); adding $\log M_{200c}$ to the class model gives an odds ratio of 0.80 per standard deviation ($p = 0.31$). The 6 isolated systems have 4 E-class satellites among 14 classified ones ($f_E = 0.286^{+0.261}_{-0.169}$), fewer than the rest of CG₄ (group-level permutation test on the per-group mean satellite f_E , $p = 0.023$) and similar to RG₄ ($p = 1.00$; Fig. F.1). With 5 contributing groups this difference is exploratory, and it cannot establish whether the absence of a larger host matters.

Group radii and crossing times use projected physical separations from the adopted Planck 2015 angular-diameter distance. This corrects the luminosity-distance convention used for that diagnostic in Paper I, decreasing the values by 3.3–9.6% (median 7.2%) without changing the reported correlations at quoted precision. The optical-colour offsets remain control dependent; full diagnostics are in the online supplement.

F.2. Continuous group-scale descriptors

As an exploratory complement to the class comparison, Table F.1 fits satellite E-versus-S classification against four continuous group-scale descriptors: $\log \langle R_{ij} \rangle$, σ_v , $f_{L,BGG}$,

and $\log t_{cross}$, adjusting for stellar mass and clustering by group. Descriptors are standardised within sample. Within CG₄, only compactness is associated with the satellite mix: smaller $\langle R_{ij} \rangle$ corresponds to more E-class satellites (OR 0.65, 95% CI [0.50, 0.84], $p_{BH} = 0.005$); the other descriptors are weak.

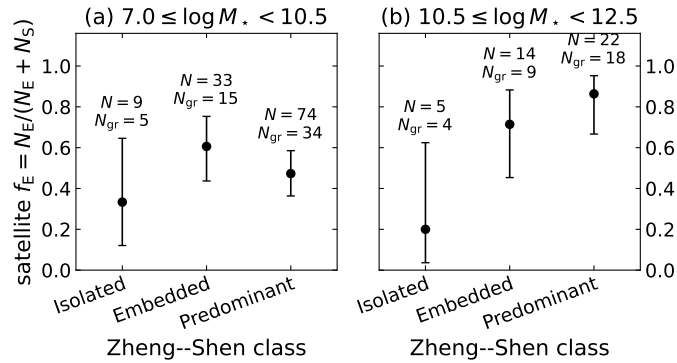


Fig. F.1. Satellite E-class fractions of the three compact-group classes in two stellar-mass bins, with Wilson 95% intervals and the numbers of classified galaxies and contributing groups; horizontal bands show the control samples.

Table F.1. Continuous group-scale screen

Descriptor	CG ₄	Control _{4B}	Control _{4C}	RG ₄
$\log\langle R_{ij} \rangle$	0.65 [0.50, 0.84] $p_{\text{BH}}=0.005$	0.80 [0.72, 0.89] $p_{\text{BH}}=1.2 \times 10^{-4}$	0.70 [0.63, 0.78] $p_{\text{BH}} < 10^{-4}$	0.92 [0.64, 1.34] $p_{\text{BH}}=0.76$
σ_v	0.90 [0.65, 1.25] $p_{\text{BH}}=0.74$	1.03 [0.93, 1.15] $p_{\text{BH}}=0.74$	1.21 [1.09, 1.35] $p_{\text{BH}}=0.002$	0.90 [0.61, 1.32] $p_{\text{BH}}=0.74$
$f_{L,\text{BGG}}$	1.19 [0.88, 1.59] $p_{\text{BH}}=0.59$	1.02 [0.90, 1.16] $p_{\text{BH}}=0.78$	1.06 [0.95, 1.19] $p_{\text{BH}}=0.59$	1.19 [0.79, 1.81] $p_{\text{BH}}=0.74$
$\log t_{\text{cross}}$	0.88 [0.66, 1.17] $p_{\text{BH}}=0.74$	0.87 [0.78, 0.97] $p_{\text{BH}}=0.03$	0.73 [0.65, 0.81] $p_{\text{BH}} < 10^{-4}$	0.99 [0.69, 1.43] $p_{\text{BH}}=0.98$

Notes. Odds ratios and 95% intervals are per within-sample standard deviation, adjusted for stellar mass. p_{BH} denotes Benjamini–Hochberg adjustment over the exploratory screen. $N = 157$ CG₄ satellites in 59 groups; control counts are given in the released table.

Appendix G: Pooled models, matching details, and selection diagnostics

G.1. Pooled adjusted models

As a compact summary across the heterogeneous control definitions we also fit binomial models to the pooled catalogue. Because the control samples overlap heavily, the pooled control side is first deduplicated by SDSS objid and standard errors are clustered by physical Lim group; the adjustment set is that of Table 4. This pooled result does not replace the per-control contrasts. For quenched status, the CG₄ odds ratio is 1.35 (95% CI [0.98, 1.86]; $p = 0.068$). For quenched satellite status, the CG₄ odds ratio is 1.53 (95% CI [1.10, 2.11]; $p = 0.011$). For GZ1 E-class morphology, the CG₄ odds ratio is 1.77 (95% CI [1.32, 2.37]; $p = 1.4 \times 10^{-4}$). For GZ1 E-class satellite morphology, the CG₄ odds ratio is 2.08 (95% CI [1.51, 2.88]; $p < 10^{-4}$). For GZ1 E-class BGG morphology, the CG₄ odds ratio is 0.78 (95% CI [0.39, 1.53]; $p = 0.464$). The pooled coefficients reinforce the satellite morphology shift; the corresponding forest plot is provided in the online supplement. The quenched estimate depends on the control sample (Sect. 3.3).

G.2. Quartet-level comparisons

Treating each quartet as the unit, we count its E-class satellites among its classified satellites. A binomial model of these counts on the CG₄ indicator and the quartet redshift, BGG stellar mass, quartet luminosity, and σ_v , fitted to all quartets with standard errors clustered by physical group, gives odds ratios of 2.14 [1.54, 2.96], 1.80 [1.30, 2.48], and 2.72 [1.56, 4.75] against Control_{4B}, Control_{4C}, and RG₄ (54 CG₄ quartets against 630, 632, and 52 control quartets). One-to-one matching on the same four variables gives the differences in satellite E-class fraction of Table G.1. The propensity model is a logistic regression on standardised covariates with the default L2 penalty of scikit-learn ($C = 1$); CG₄ quartets are matched greedily, in order of their closest available control, within a caliper of 0.2 standard deviations of the propensity logit, which retains 54 of the 630 Control_{4B} and 54 of the 632 Control_{4C} quartets, and 28 of the 52 RG₄ quartets, with maximum absolute standardised mean differences of 0.17, 0.09, and 0.25 after matching. The matched differences depend on these choices. With an unpenalised propensity model they become 0.15, 0.14, and 0.26; adding the mean satellite stellar mass to the match gives

Table G.1. Matched quartets: satellite E-class fractions

Control	Pairs	Δ	95% CI	p_{perm}
Control _{4B}	54	0.160	[0.034, 0.284]	0.02
Control _{4C}	54	0.114	[−0.015, 0.238]	0.08
RG ₄	28	0.196	[0.036, 0.351]	0.03
Pooled	54	0.167	[0.040, 0.290]	0.02

Notes. Δ is the matched mean E-class satellite-fraction difference (CG₄ minus control), with 95% pair-bootstrap intervals (9999 draws); p_{perm} is the paired sign-flip permutation p-value. The final row uses the deduplicated control pool.

0.20, 0.09, and 0.26; random processing orders move the RG₄ value between 0.17 and 0.28 (5th–95th percentiles); perturbing σ_v in both samples by a further realisation of the four-member gapper noise moves the Control_{4C} value between 0.08 and 0.22, with a permutation $p < 0.05$ in 62% of the realisations; and without σ_v the Control_{4B} value becomes 0.28 because a different set of control quartets is selected, dynamically cooler than their CG₄ partners (median σ_v of 129 against 160 km s^{−1}), while drawing the Control_{4B} controls from the deduplicated pool gives 0.10. Because the match uses a small fraction of the control quartets and is sensitive to its implementation, we treat it as a robustness check of the all-quartet comparison rather than as a separate test.

G.3. Galaxy-level matching details

The galaxy-level control pool is deduplicated by SDSS objid before matching: 5828 control rows collapse to 3609 unique galaxies. The propensity model (penalised as for the quartets) uses $\log M_*$, redshift, luminosity rank within the quartet, quartet luminosity, and σ_v , and the match is one-to-one without replacement within exact rank strata. For the 234 matched CG₄ galaxies, the matched E-class fraction difference is 0.14 (95% interval [0.04, 0.24]), and 0.19 for the 130 satellite pairs alone; the quenched-fraction difference is 0.03 ([−0.06, 0.12]). The 14 unmatched CG₄ galaxies lay outside the common propensity-score support or exceeded the caliper. The maximum absolute covariate standardised mean difference falls from 0.37 to 0.08 (balance plot in the online supplement). By kept label, the matched controls comprise 159 Control_{4B}, 57 Control_{4C}, 18 RG₄; physically, 18 of the 234 matched controls are RG₄ galaxies (which also

appear in Control_{4B} through the sample overlap), and the per-galaxy origin of every matched control is released with the derived tables. Confidence intervals and p-values come from a bootstrap blocked by the treated galaxy's compact group.

The 234 matched controls occupy 204 physical Lim groups, and 26 control hosts link more than one treated CG₄ group. Resampling the 33 connected components of the treated-group/control-host graph instead, the E-class fraction difference remains 0.139 (95% interval [0.051, 0.244]; $p = 0.004$), and the quenched-fraction difference is 0.031 ([−0.090, 0.120]; $p = 0.649$). The analogous matched-size sensitivities are −0.019 dex for the Simard radius ($p = 0.310$), −0.033 dex for the Petrosian radius ($p = 0.016$), and −0.008 for the concentration ($p = 0.896$).

G.4. Selection diagnostics

The fractions with complete measurements for the major derived quantities, including the Simard and Petrosian sizes and the $D_n4000/H\delta_A$ indices, are shown in the online supplement. No major availability difference has both an adjusted p-value below 0.05 and a difference exceeding 10 percentage points.

The adjustment set contains no group-mass covariate. The group tables do carry an M_{200c} estimate, but it is the Yang–Lim abundance-matching M_{180m} of the identified *host* halo (converted as in Paper I, Appendix A), not a quartet property: for embedded quartets it describes the richer host while for isolated systems it describes the quartet itself, it derives from group luminosity by abundance matching and is therefore nearly collinear with the group-luminosity covariate, and it is entangled with the compact/loose selection. We therefore exclude it from the covariate set. A sensitivity fit adding $\log M_{200c}$ to the adopted covariates leaves every E-class conclusion unchanged: the pooled odds ratio moves from 1.77 to 1.62 ($p = 0.0020$), and the per-control odds ratios become 2.00, 1.56, and 2.65 against Control_{4B}, Control_{4C}, and RG₄ (all $p \leq 0.004$).

Restricting the pairwise tidal index of Sect. 3.2 to quartet co-members is innocuous: recomputing T_i with all spectroscopic Lim members of the identified host group included (positions from the group catalogue; stellar masses from MPA–JHU where available, so the host-inclusive index is itself a lower bound) shifts $\log T_i$ by a median of 0.009 dex (sample medians ≤ 0.02 dex, and identically zero for RG₄, whose groups have exactly four members), with rank correlation 0.91 between the two definitions, and refitting the with-index E-class model on the host-inclusive index moves the attenuated CG₄ coefficient only from 1.39 to 1.33. T_i is thus the local, galaxy-scale tidal density traced by catalogued galaxies; any smooth host-halo contribution is probed separately through the host-mass sensitivity fit above and the within-host analysis of Sect. 3.2. The colour-selection mass diagnostics are provided in the online supplement.

Appendix H: Robustness details

Sample-definition sensitivities. Restoring the 16 Zheng–Shen split systems (78 compact groups) while retaining the model definitions of Table 4 leaves the morphology result qualitatively unchanged: the adjusted all-galaxy E-class odds ratios against Control_{4B}, Control_{4C}, and RG₄ are

1.97 [1.46, 2.64], 1.49 [1.12, 1.96], and 2.62 [1.69, 4.06], and the corresponding satellite odds ratios are 2.57 [1.88, 3.50], 1.84 [1.38, 2.46], and 2.96 [1.75, 5.01]. Retaining the Lim 3688 quartets, whose outlying member ($z = 0.048$, against 0.032–0.034 for the other 6 members) gives them $\sigma_v = 1979$ and 2042 km s^{-1} , likewise leaves the coefficients essentially unchanged: the all-galaxy and satellite odds ratios are 2.18 and 2.65 against Control_{4B}, and 1.68 and 1.94 against Control_{4C}.

Crowding. Galaxy Zoo morphology is image-based and SDSS spectroscopy loses close pairs to fibre collisions, so crowding could bias both the votes and the spectroscopic measurements. The flag of Sect. 3.5 marks galaxies with another galaxy of the Lim et al. (2017) catalogue, at any redshift, within 55 arcsec; it is set for 27, 14, 16, and 10% of the CG₄, Control_{4B}, Control_{4C}, and RG₄ members, against 19, 6, 10, and 5% when only the other quartet members are considered. Without the flagged galaxies the E-class fraction $N_E/(N_E + N_S)$ changes from 0.590 to 0.577 in CG₄ and from 0.476 to 0.456 in Control_{4C}, and the satellite odds ratios become 2.54, 1.91, and 2.88. With the flag as a covariate instead, the all-galaxy CG₄ odds ratios are 2.01, 1.55, and 2.59, while flagged galaxies are themselves more often in the E class (odds ratios 1.61–1.91). The Lim catalogue is spectroscopic, so companions that never received a redshift are absent and the flag cannot recover galaxies lost to fibre collisions; the test shows that the morphology contrast does not come from the catalogued close pairs, not that crowding is irrelevant.

Without σ_v . Refitting the per-control and pooled models without the σ_v covariate, on identical samples and with identical clustering, moves the E-class odds ratios from 2.14 to 2.22 (Control_{4B}), 1.67 to 1.68 (Control_{4C}), and 2.81 to 2.76 (RG₄), less than their standard errors. The galaxy-level match retains the same 234 CG₄ galaxies with an E-class fraction difference of 0.16; the quartet-level matches are discussed in Appendix G.2.

Projected position. Table H.1 collects the satellite E-class odds ratios for the radial specifications of Sect. 3.2, between samples and within hosts. Between samples, scaling the projected distance by r_{200} of the host halo, replacing $\log R_{\text{BGG}}$ by a restricted cubic spline with knots at its 10th, 50th, and 90th percentiles, or restricting the controls to satellites no farther from their BGG than 319 kpc, the largest CG₄ distance, changes the position-adjusted odds ratios by less than their uncertainties; the corresponding differences in E-class fraction at common distance are 0.13, 0.08, and 0.14 for the spline and 0.13, 0.08, and 0.13 on the common support. Within hosts, the estimate depends on how distance enters: it is close to unity with a logarithmic or spline term for the satellites, higher with a linear distance term for all members, and below unity when the co-members are restricted to the distances of the members, where only 501 galaxies remain. This dependence on the specification is itself a symptom of the small radial overlap.

Within-host estimators. The conditional-logit sample of all host members contains 791 galaxies in 53 informative host

Table H.1. Satellite E-class odds ratios with projected position

Specification	Control _{4B}	Control _{4C}	RG ₄
<i>Between samples</i>			
No position term	2.63 [1.88, 3.68]	1.93 [1.40, 2.66]	3.03 [1.72, 5.34]
$\log R_{\text{BGG}}$	1.80 [1.28, 2.55]	1.39 [1.01, 1.92]	1.95 [1.08, 3.53]
$\log(R_{\text{BGG}}/r_{200})$	1.70 [1.19, 2.42]	1.34 [0.97, 1.87]	1.87 [1.01, 3.45]
Spline in $\log R_{\text{BGG}}$	1.78 [1.26, 2.52]	1.40 [1.01, 1.95]	1.93 [1.00, 3.71]
Common support	1.77 [1.25, 2.53]	1.43 [1.03, 1.99]	1.90 [0.94, 3.83]
<i>Within hosts</i>			
All, no distance	1.90 [1.23, 2.95]		
All, linear R	1.47 [0.92, 2.35]		
Satellites, no distance	1.89 [1.21, 2.95]		
Satellites, $\log R$	0.93 [0.55, 1.57]		
Satellites, spline	0.85 [0.49, 1.46]		
Common support	0.74 [0.41, 1.33]		

Notes. Between samples: odds ratios of the CG₄ indicator in the satellite models of Table 4; the common-support row keeps control satellites within the largest CG₄ distance and includes $\log R_{\text{BGG}}$. Within hosts: conditional-logit odds ratios of CG₄ membership at fixed stellar mass (and host luminosity rank, except in the spline and common-support rows), stratified by host; the common-support row keeps co-member satellites within the 95th percentile of the member distances (784 kpc) and includes $\log R$.

strata. A binomial model with host fixed-effect dummies and cluster-robust standard errors, with the linear distance, gives an odds ratio of 1.57 (95% CI [1.02, 2.43]), close to the corresponding conditional-logit value of Table H.1. For quenched status, the member odds ratio is 1.22 [0.76, 1.94] without a distance term and 0.48 [0.27, 0.84] for satellites with $\log R$. A host-BGG alignment check supports the satellite interpretation: 56 of 62 non-split CG₄ systems have the same quartet and Lim-host BGG, the 6 misaligned systems are embedded systems whose host BGG is brighter than the compact-group BGG, and all 186 quartet-rank CG₄ satellites are Lim-host satellites; restricting CG₄ to the 56 BGG-aligned systems leaves the three satellite E-class contrasts stable (against Control_{4C}, OR 1.98).

DS18 classification. The DS18 cross-match of Sect. 2.2 covers 173/248 (70%) of the CG₄ galaxies and 2263/3609 (63%) of the deduplicated controls, i.e. the DR7-era population; it is not worse in CG₄, but neither rate establishes representativeness. With the DS18 $T\text{Type} \leq 0$ outcome and the covariates, standardisation, and physical-group clustering of Table 4, the CG₄ coefficients are positive in all three contrasts (Table 6); support is broad except that 83% of CG₄ values lie within the RG₄ group-luminosity range, limiting that smallest comparison. Because DS18 P_{S0} separates S0 from E only within $T\text{Type} \leq 0$, the environment dependence of the E/S0 mix is left to a dedicated analysis.

Appendix I: Extended size diagnostics

The size-availability counts are shown in the online supplement; the descriptive mass-size relations and the per-sample sizes predicted at $\log(M_{\star}/M_{\odot}) = 10.3$ are provided in the online supplement. The concentration index $C = R_{90,r}/R_{50,r}$ provides a model-independent structural proxy: the adjusted satellite coefficient is 0.068 ($p = 0.124$), and the matched concentration difference is -0.008 ($p = 0.904$), offering no evidence for a fixed-mass concentration shift.

The crowding checks mirror Appendix H: excluding galaxies with a projected neighbour within 55 arcsec leaves a satellite coefficient of -0.017 dex ($p = 0.340$), and adding the close-neighbour flag as a covariate gives -0.009 dex ($p = 0.559$).

The per-galaxy difference between the two measures, $d = \log R_{e,r} - \log R_{50,r}$, averages 0.054 dex, and is larger for GZ1 E-class galaxies (0.074 dex) than for S-class galaxies (0.042 dex), as expected if the Petrosian aperture preferentially truncates concentrated profiles. Its rank correlation with the nearest-neighbour separation is $\rho_s = -0.051$ ($p = 0.006$), so residual blending does not drive a strong measure-dependent size systematic.

Adding the pairwise projected tidal index of Sect. 3.2 to the all-galaxy size model changes the CG₄ coefficient from -0.012 dex to 0.005 dex on the pairwise complete-case sample; the local projected-density proxy attenuates the size coefficient in the same way it attenuates the morphology coefficient.

The radial diagnostic is exploratory: mass- and redshift-adjusted satellite size residuals correlate with projected BGG-centric distance with $\rho_s = 0.17$ ($p = 0.026$) in CG₄ and $\rho_s = 0.18$ ($p < 10^{-4}$) in the pooled controls. A more negative inner-region residual would be the tidal-truncation expectation; the observed trends are similar in CG₄ and the controls, so the data do not isolate a CG₄-specific inner-region size suppression.

Within morphology classes, the satellite-only fits give 0.036 dex for GZ1 E-class satellites ($p = 0.068$) and -0.005 dex for S-class satellites ($p = 0.831$), and the CG₄×morphology interaction is 0.032 dex ($p = 0.348$). These strata are secondary diagnostics of whether size carries information beyond the morphology shift.